

PHYS 201 - Quantum Mechanics

Homework #4: Wave Functions

Instructions: Answer all questions. Show your work. Submit as PDF.

1. Derive the time-independent Schrödinger equation from first principles.
2. For a particle in an infinite square well of width L :
 - a) Find the normalized wave functions.
 - b) Calculate the energy eigenvalues.
 - c) Plot the probability density for $n=1,2,3$.
3. Explain the physical significance of the Born interpretation.
4. Prove that the momentum operator is Hermitian.

Due: April 20, 2026 at 11:59 PM via Gradescope.

Late submissions will lose 10% per day.
